boAt

boAt Lifestyle is an Indian consumer electronics brand specializing in stylish, affordable audio devices and wearable technology. It focuses on delivering innovative, youth-centric products that combine performance, design, and value to enhance everyday lifestyles.

boAt Sales Performance Analytical Report

Power BI Dashboard Analysis of Revenue, Channel Mix, Product Category, Discount Behaviour and Customer Segments | 2024

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Executive Summary

This report consolidates the Power BI dashboard analysis built on the boAt sales dataset, covering revenue, order quantity, discounting, profitability, sales channel, product category and customer-segment dimensions. Total revenue across the year came in at approximately Rs 21.84 Million on 10,000 orders with an average quantity of 1.74 units per order (as summarised in the KPI cards). Total profit stood at Rs 10.42 Million, yielding a healthy overall profit margin of 47.7%. Revenue performance diverged sharply by month, with festival months (October-November) significantly outperforming the April-June lean period. Online channels dominate the mix, with Amazon and Flipkart together accounting for more than 61% of total revenue. On the product side, Earbuds is the clear volume leader with 3,001 orders - nearly double the next largest category (Neckbands at 2,210 orders).

Rs 21.84M	Rs 10.42M	10,000
TOTAL REVENUE	TOTAL PROFIT	TOTAL ORDERS

4.33 *	1.74	47.7%
AVG RATING	AVG SALES (QTY)	PROFIT MARGIN

Purpose of the Report

The purpose of this report is to demonstrate practical proficiency in Python (Data Generation & Cleaning), SQL (Data Analysis) and Microsoft Power BI (Dashboard & Visualization) by analyzing a simulated boAt sales dataset. The project showcases skills in data cleaning, transformation, DAX measures, interactive dashboard creation, and business insight generation, illustrating the ability to convert raw data into meaningful visualizations and actionable insights. This report has been prepared solely for learning, practice, and portfolio purposes.

Data

Dataset Description, Cleaning & Transformation

Dataset Description

The dataset used in this project is a synthetically generated dataset created for educational and analytical purposes. It simulates the operations of a retail and e-commerce business similar to boAt by incorporating realistic information related to customers, products, sales transactions, sales channels, and delivery operations. Although the data closely resembles real-world business scenarios, it does not contain any actual customer information or confidential company data.

Attribute	Detail
Total Records	10,000 orders
Time Period	January - December 2024
Products	37 real boAt products
Categories	6 (Earbuds, Headphones, Neckband, Smartwatch, Speaker, Wired Earphones)
States/UTs	35 Indian states and union territories
Cities	139 cities across India
Columns	27 attributes per record

Data Cleaning

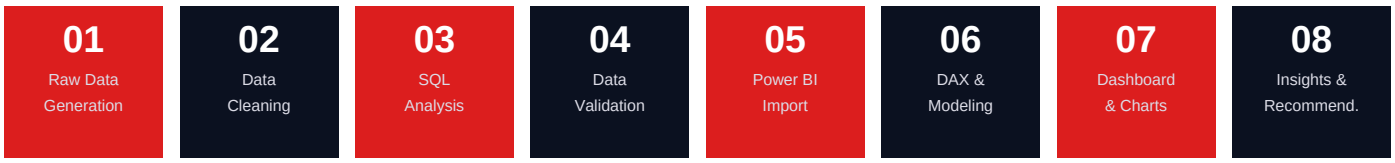
- Checked missing values - No null values found
- Checked duplicate records - 0 duplicate Order IDs
- Parsed Order_Date to proper datetime format (DD-MM-YYYY)
- Validated no negative revenues in dataset
- Ratings clipped to valid [1.0, 5.0] range
- Removed inconsistencies in data types

Data Transformation

The raw dataset was generated using Python (generate_boat_data.py) with realistic weighted distributions for categories, sales channels, geography, and seasonality. The cleaning pipeline (data_cleaning.py) enriched the dataset with additional calculated columns including Profit_Margin, Quarter, Quarter_Label, Day_of_Week, and Price_Segment using Pandas transformations. The final cleaned CSV (boat_sales_cleaned.csv) was imported into Power BI, where DAX measures were created for KPI cards, formatted displays, and conditional aggregations.

Methodology

The analysis followed a structured and systematic workflow, beginning with raw data generation and progressing through cleaning, SQL validation, and Power BI visualization to generate actionable business recommendations. Each stage built upon the previous one, ensuring that every insight presented in the report is accurate, transparent, and fully traceable.



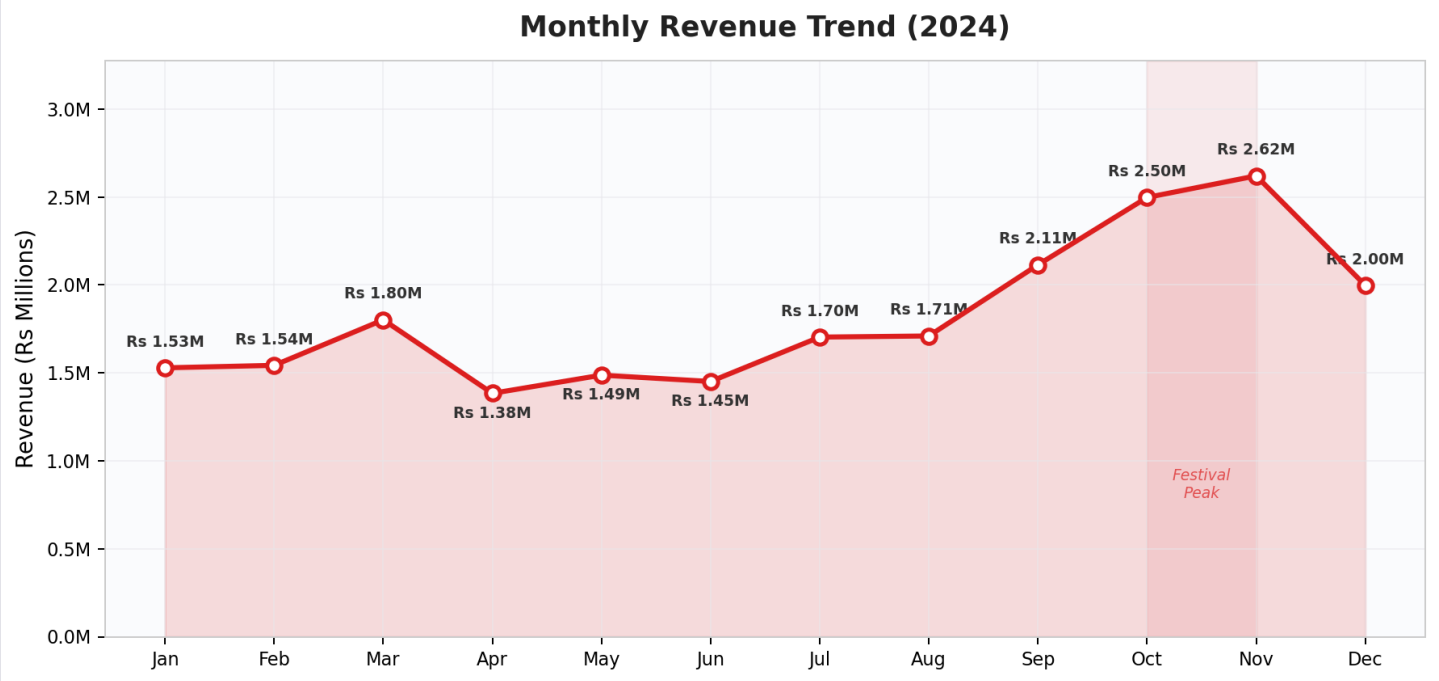
Stage	Description
Raw Data	Generated 10,000 realistic sales records using Python with weighted distributions
Data Cleaning	Missing values, date formats, data types corrected; calculated columns added
Data Integration	Profit Margin, Quarter, Price Segment, Day of Week derived from base columns
Data Validation	Cross-checked SQL query totals against Power BI DAX measures for accuracy
Power BI Import	Loaded cleaned CSV, created Date Table, established relationships
DAX Measures	Created KPI measures, formatted displays, and conditional aggregations
Dashboard	Built interactive dashboard with 9 visual types, 3 slicers, and dark theme
Insights	Interpreted each visual to surface findings and business recommendations

Analysis

Monthly Revenue Trend

January - December 2024

Monthly revenue follows a clear seasonal pattern shaped by India's festival calendar and e-commerce sale events. The year begins with moderate revenue in January (Rs 1.53M) and February (Rs 1.54M), picks up in March (Rs 1.80M), then dips to its lowest point in April (Rs 1.38M) - marking the beginning of the lean quarter. May and June remain subdued at Rs 1.49M and Rs 1.45M respectively.



The mid-year recovery begins in July (Rs 1.70M), coinciding with Amazon Prime Day sales, and continues through August (Rs 1.71M) driven by Independence Day promotions. September marks the inflection point (Rs 2.11M) as pre-festival demand kicks in. The peak arrives in October (Rs 2.50M) and November (Rs 2.62M) - fuelled by Dussehra, Navratri, Diwali, and major e-commerce festivals.

Month	Orders	Revenue (Rs)	Profit (Rs)
January	692	1.53M	0.76M
February	671	1.54M	0.77M
March	769	1.80M	0.90M
April	611	1.38M	0.69M
May	603	1.49M	0.74M
June	635	1.45M	0.72M
July	783	1.70M	0.79M
August	793	1.71M	0.78M
September	896	2.11M	1.06M
October	1,269	2.50M	1.08M

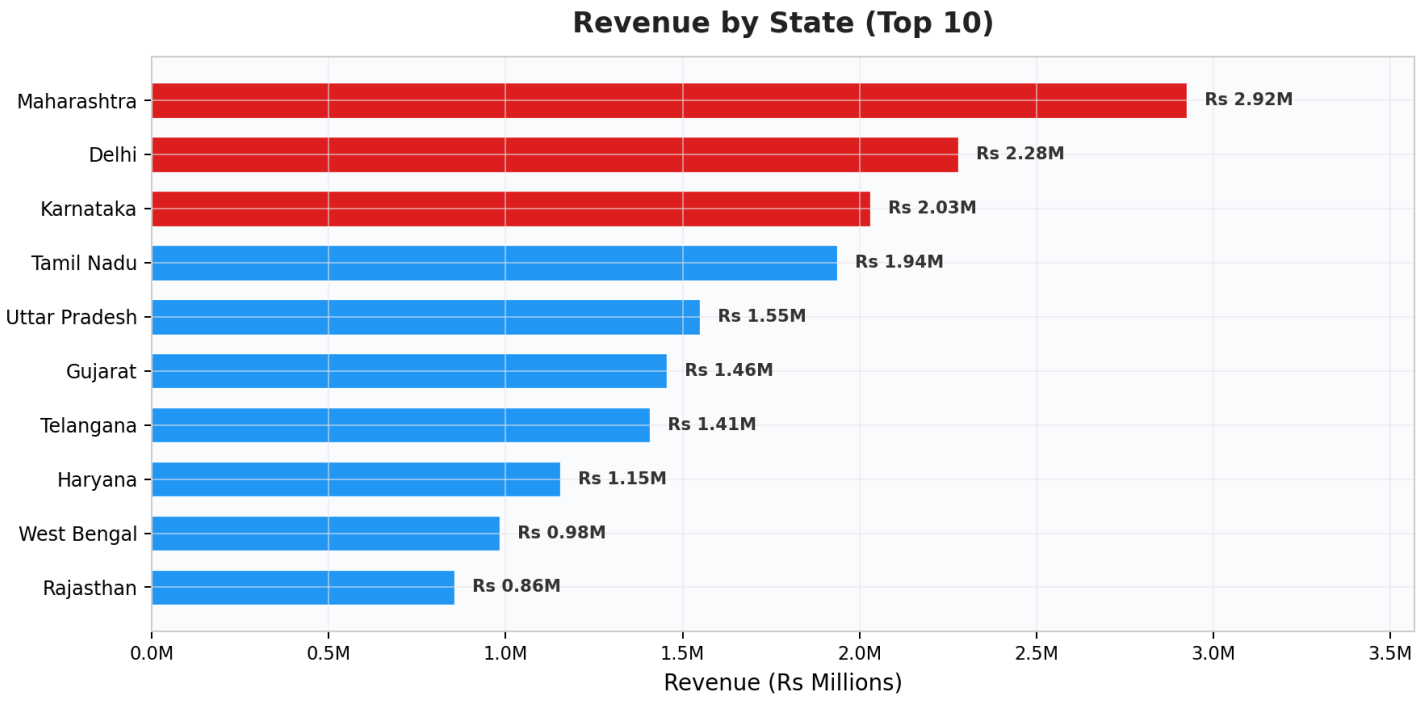
Month	Orders	Revenue (Rs)	Profit (Rs)
November	1,351	2.62M	1.14M
December	927	2.00M	1.00M

Key Insight: *November is the peak revenue month at Rs 2.62M, contributing 12.0% of annual revenue. The Oct-Nov festive window alone generates Rs 5.12M (23.4%) of the full year's revenue.*

Revenue by State

Map Analysis - Top 10 States

State-wise sales are concentrated in western and southern India, with Maharashtra emerging as the strongest revenue contributor at Rs 2.92M (13.4% of total), followed by Delhi at Rs 2.28M (10.4%) and Karnataka at Rs 2.03M (9.3%). Tamil Nadu (Rs 1.94M) and Uttar Pradesh (Rs 1.55M) complete the top five.

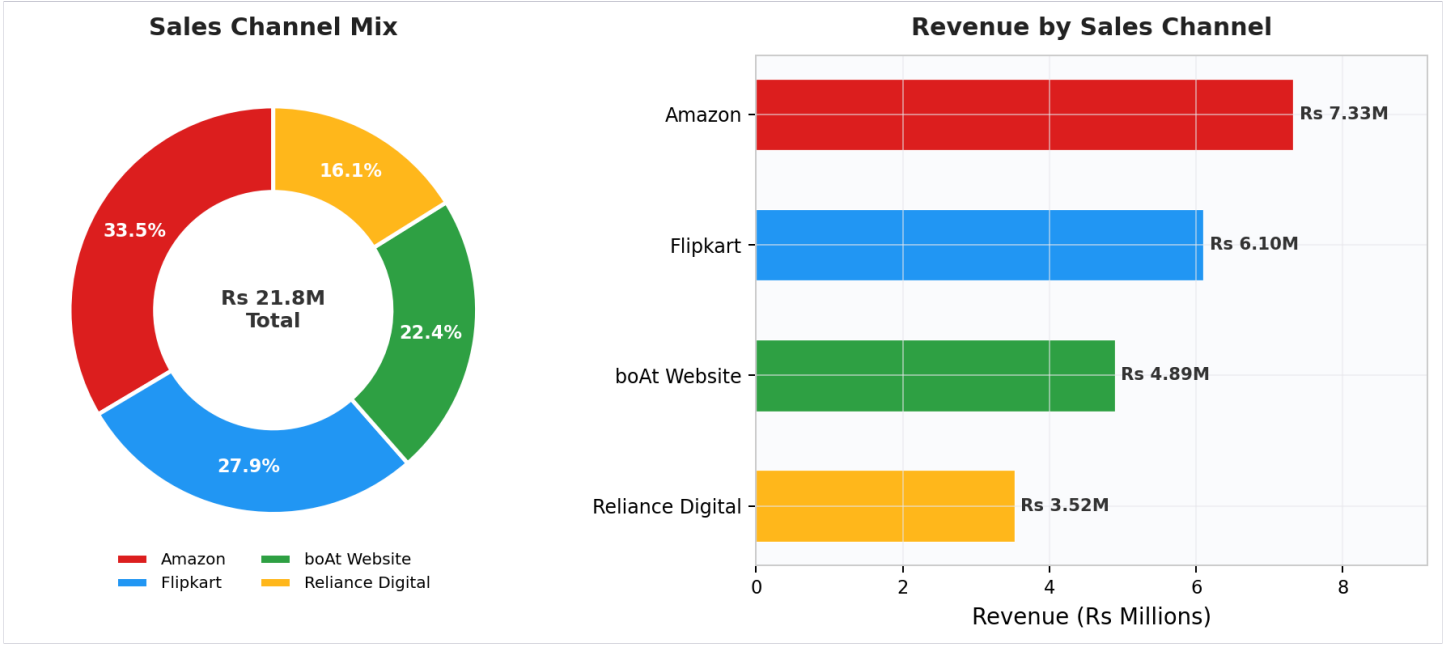


Rank	State	Region	Orders	Revenue (Rs)	Profit (Rs)
1	Maharashtra	West	1,357	**2.92M**	1.40M
2	Delhi	North	1,035	**2.28M**	1.09M
3	Karnataka	South	914	**2.03M**	0.97M
4	Tamil Nadu	South	888	**1.94M**	0.92M
5	Uttar Pradesh	North	728	**1.55M**	0.74M
6	Gujarat	West	647	**1.46M**	0.69M
7	Telangana	South	636	**1.41M**	0.68M
8	Haryana	North	507	**1.15M**	0.55M
9	West Bengal	East	454	**0.98M**	0.46M
10	Rajasthan	North	381	**0.86M**	0.41M

Key Insight: *The Top 5 states alone contribute 49.0% of total revenue, indicating high geographic concentration. The Northeast region (8 states, 310 orders, Rs 0.62M) represents a significant untapped market.*

Sales Channel & Retail Platform Performance

Online channels are the company's primary revenue driver. Amazon (Rs 7.33M) and Flipkart (Rs 6.10M) are the two largest revenue contributors, together accounting for 61.5% of total sales. The boAt Official Website adds Rs 4.89M (22.4%), reinforcing the importance of its direct-to-consumer (D2C) channel. Reliance Digital contributes Rs 3.52M (16.1%) as the offline retail partner.

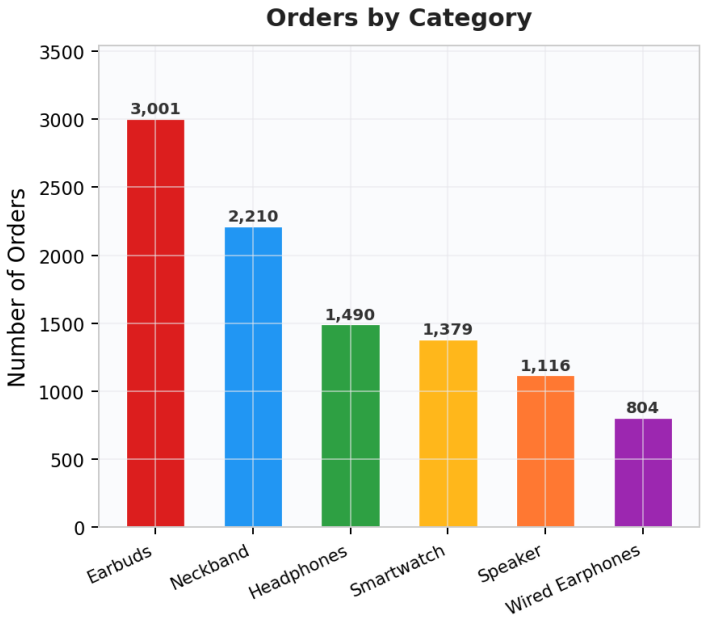


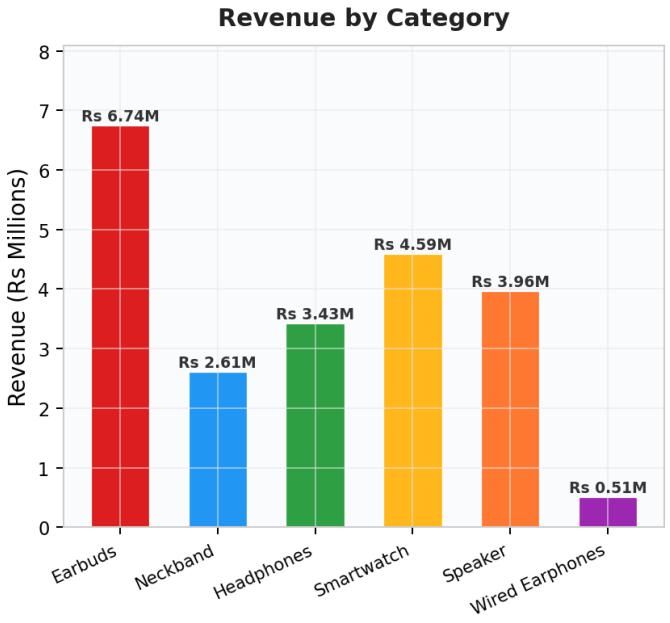
Sales Channel	Orders	Revenue (Rs)	Profit (Rs)	Avg Rating	Share %
Amazon	3,501	7.33M	3.32M	4.34	33.6%
Flipkart	2,976	6.10M	2.70M	4.33	27.9%
boAt Website	2,012	4.89M	2.54M	4.32	22.4%
Reliance Digital	1,511	3.52M	1.87M	4.34	16.1%

Key Insight: *While Amazon leads in revenue (Rs 7.33M), the boAt Website delivers the highest profit per order (Rs 1,260) compared to Amazon (Rs 947) - suggesting D2C channel expansion could significantly boost profitability.*

Product Category Performance

Earbuds is the runaway volume leader, generating 3,001 orders - 30% of the total - and commanding the highest revenue at Rs 6.74M. Smartwatches rank second in revenue (Rs 4.59M) despite fewer orders (1,379), driven by higher average selling prices (Rs 1,908). Speakers contribute Rs 3.96M from 1,116 orders with the highest category rating (4.43).



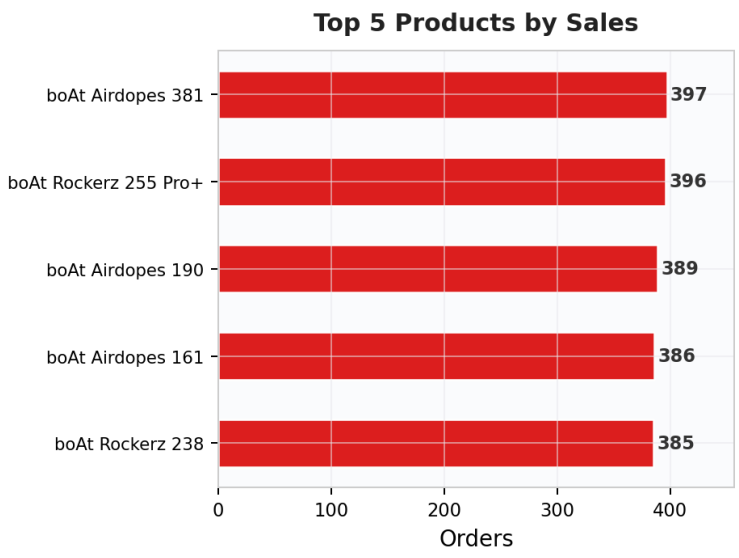


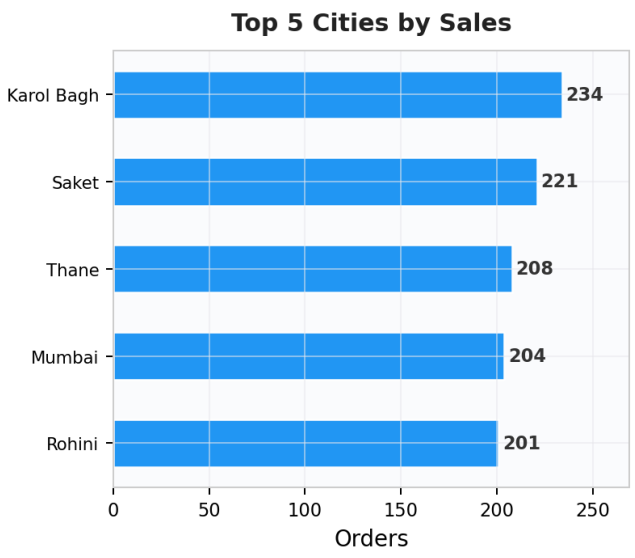
Category	Orders	Revenue (Rs)	Profit (Rs)	Avg Rating	Avg Price (Rs)
Earbuds	3,001	6.74M	3.23M	4.40	1,266
Smartwatch	1,379	4.59M	2.16M	4.09	1,908
Speaker	1,116	3.96M	1.87M	4.43	2,050
Headphones	1,490	3.43M	1.65M	4.36	1,345
Neckband	2,210	2.61M	1.25M	4.37	681
Wired Earphones	804	0.51M	0.25M	4.24	367

Key Insight: *Earbuds should remain the primary focus for inventory planning, marketing spend and new SKU launches, while the Smartwatch category represents the strongest secondary growth lever with the highest average selling price.*

Top 5 Products & Cities by Sales Volume

The boAt Airdopes 381 is the top seller with 397 orders, followed closely by boAt Rockerz 255 Pro+ (396) and boAt Airdopes 190 (389). The top 5 products are all within a tight 12-unit range (385-397), indicating a healthy distribution of demand. Sales are heavily skewed toward Delhi, which claims 3 of the top 5 city slots.





Rank	Product	Orders	Revenue (Rs)
1	boAt Airdopes 381	397	1.14M
2	boAt Rockerz 255 Pro+	396	0.51M
3	boAt Airdopes 190	389	0.78M
4	boAt Airdopes 161	386	0.77M
5	boAt Rockerz 238	385	0.38M

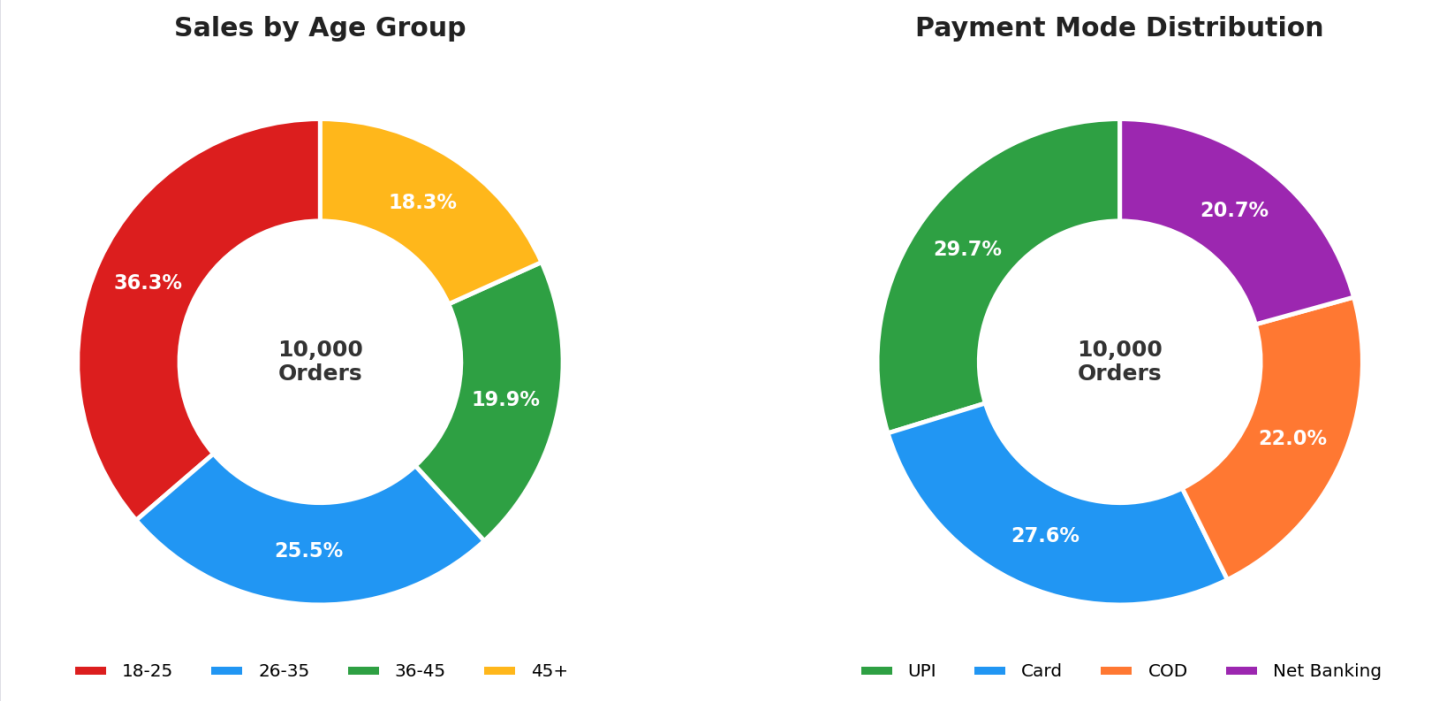
Key Insight: *While Airdopes 381 and Rockerz 255 Pro+ have nearly identical order volumes, the Airdopes 381 generates 2.2x more revenue (Rs 1.14M vs Rs 0.51M) due to its higher unit price - reinforcing that premium Earbuds models drive disproportionate value.*

Rank	City	State	Orders	Revenue (Rs)
1	Karol Bagh	Delhi	234	0.47M
2	Saket	Delhi	221	0.48M
3	Thane	Maharashtra	208	0.45M
4	Mumbai	Maharashtra	204	0.44M
5	Rohini	Delhi	201	0.48M

Key Insight: *Despite Karol Bagh having the highest order count, Rohini and Saket deliver higher revenue per order - indicating customers in these areas purchase higher-value products.*

Sales by Age Group & Payment Modes

Out of 10,000 total orders, the 18-25 age group forms the largest segment at 36.3% (3,630 orders), followed by 26-35 at 25.5%. The dominance of younger demographics - with the 18-35 bracket accounting for 61.8% of all orders - is consistent with boAt's youth-centric brand positioning. UPI is the most preferred payment method at 29.7%.



Age Group	Orders	Share %
18-25	3,630	36.3%
26-35	2,552	25.5%
36-45	1,991	19.9%
45+	1,827	18.3%

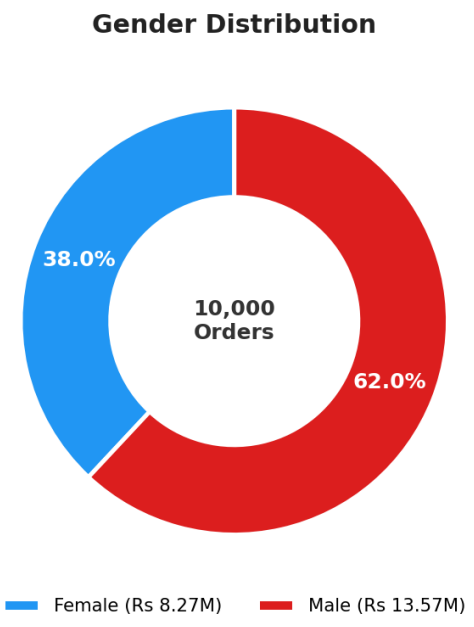
Key Insight: *The 18-25 segment is the brand's core audience. Marketing campaigns, influencer partnerships, and student-offer bundles should be heavily targeted at this cohort to maximize conversion.*

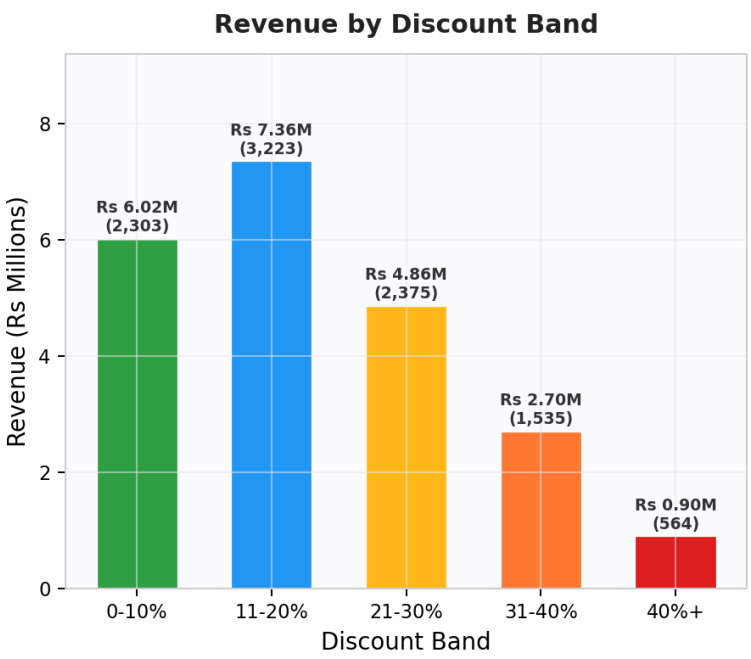
Payment Mode	Orders	Share %
UPI	2,974	29.7%
Card	2,757	27.6%
COD	2,198	22.0%
Net Banking	2,071	20.7%

Key Insight: *UPI's leadership reflects India's digital payment revolution. The relatively high COD share (22%) presents an opportunity to incentivize prepaid orders with small discounts, reducing return/cancellation rates typical of COD orders.*

Gender Distribution & Discount Band Analysis

Male customers account for 62.0% of all orders (6,196 orders, Rs 13.57M revenue), while female customers contribute 38.0% (3,804 orders, Rs 8.27M revenue). The 11-20% discount band is the volume sweet spot, driving the highest revenue at Rs 7.36M from 3,223 orders.





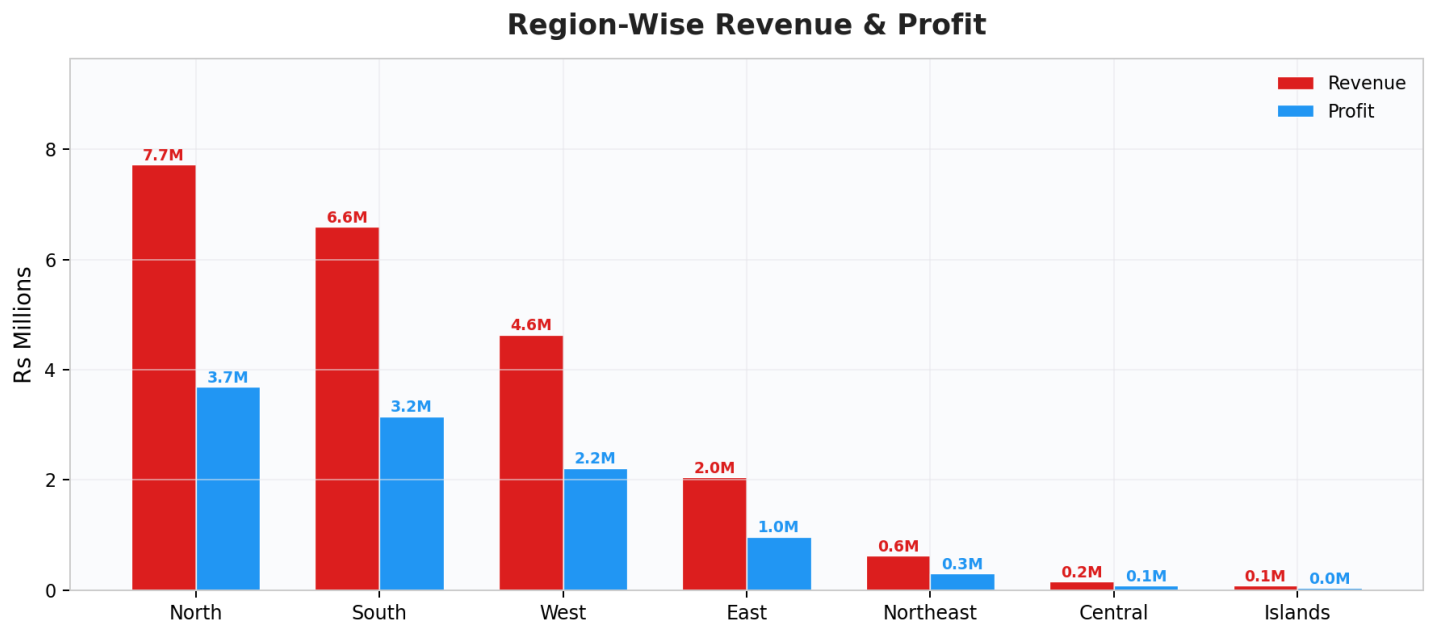
Gender	Orders	Share %	Revenue (Rs)
Male	6,196	62.0%	13.57M
Female	3,804	38.0%	8.27M

Discount Range	Orders	Avg Selling Price (Rs)	Revenue (Rs)	Avg Rating
0-10%	2,303	1,497	6.02M	4.34
11-20%	3,223	1,327	7.36M	4.33
21-30%	2,375	1,150	4.86M	4.34
31-40%	1,535	1,020	2.70M	4.32
40%+	564	886	0.90M	4.34

Key Insight: *The 11-20% discount band is the optimal sweet spot, balancing maximum order volume with strong revenue generation. Discounts beyond 30% erode margins significantly without proportional volume gains. Customer ratings remain consistent (4.32-4.34) regardless of discount level, proving that discounts do not improve customer satisfaction.*

Region-Wise Performance

North India is the largest revenue contributor at Rs 7.72M from 3,536 orders across 11 states and 52 cities. The South follows closely at Rs 6.59M (2,993 orders), while West India generates Rs 4.63M from just 3 states. The East, Northeast, Central, and Island regions collectively contribute only Rs 2.90M - representing significant growth headroom.



Region	States	Cities	Orders	Revenue (Rs)	Profit (Rs)
North	11	52	3,536	**7.72M**	3.69M
South	6	27	2,993	**6.59M**	3.15M
West	3	16	2,125	**4.63M**	2.21M
East	4	20	927	**2.05M**	0.97M
Northeast	8	19	310	**0.62M**	0.30M
Central	1	4	74	**0.15M**	0.07M
Islands	2	2	35	**0.08M**	0.03M

Key Insight: *The Northeast has 8 states but generates only Rs 0.62M - less than a single top city like Karol Bagh. This region presents the biggest untapped opportunity for distribution expansion and brand awareness campaigns.*

Key Insights & Recommendations

Cap Discounting at 30%

The 11-20% discount band is the profit-volume sweet spot, generating Rs 7.36M at maximum order volume. Beyond 30%, margins erode rapidly without proportional volume gains. Restrict deep discounts to end-of-season inventory clearance only.

Double Down on Earbuds

Earbuds drive 30% of orders and command the highest revenue (Rs 6.74M) with a strong 4.40 rating. Prioritize this category for inventory planning, marketing spend, and new SKU launches, with Smartwatches as the strongest secondary growth lever.

Expand the D2C Channel (boAt Website)

While the boAt Website has fewer orders than Amazon/Flipkart, it delivers the highest profit per order (Rs 1,260 vs Rs 947 on Amazon). Investing in the website's user experience, exclusive product launches, and loyalty programs can significantly boost margins.

Strengthen the Amazon-Flipkart Partnership

Amazon and Flipkart together deliver 61.5% of revenue. Continue investing in sponsored ads, deal-of-the-day placements, and festival-season promotions on these platforms to maintain market share.

Capitalize on the Oct-Nov Festival Window

The October-November festive period generates Rs 5.12M (23.4% of annual revenue) in just 2 months. Plan inventory build-up, marketing campaigns, and exclusive product launches around Dussehra, Diwali, and e-commerce sale events.

Target the 18-25 Segment Aggressively

This cohort drives 36.3% of all orders. Deploy influencer marketing, campus ambassador programs, student discounts, and social media campaigns tailored to Gen Z preferences to maintain and grow this base.

Expand into Underserved Regions

The Northeast (8 states, Rs 0.62M), Central (Rs 0.15M), and Islands (Rs 0.08M) collectively represent massive untapped potential. Partner with local retailers, strengthen logistics infrastructure, and run region-specific awareness campaigns.

Incentivize Prepaid Payments over COD

COD orders (22% share) typically carry higher return and cancellation rates. Offer small prepaid discounts (Rs 50-100 off) or exclusive offers for UPI/Card payments to shift the mix toward digital payments.

Dashboard Preview

Power BI Sales Analytics Dashboard

The interactive Power BI dashboard was designed with a dark navy theme (#0B1120) for visual impact and readability. It features 5 KPI cards, 3 interactive slicers, a state-wise performance map, donut charts for age group and payment modes, column charts for sales channels, an area chart for monthly revenue trend, and horizontal bar charts for top 5 products and cities.



Design: Dark navy theme (#0B1120) with card backgrounds (#131B2E), boAt red accent (#FF0000), white typography, and glassmorphic card borders. The dashboard supports interactive filtering by State, Sales Channel, and Category.

Tools Used

Python (Data Generation & Cleaning) | SQL / SQLite (Data Analysis) | Microsoft Power BI (Dashboard & DAX)

Pandas, CSV, Random libraries | DAX Measures | Interactive Visualizations

Dataset: 10,000 orders | 37 products | 6 categories | 35 states | 139 cities | Year 2024